

Problem Set 1
Intermediate Microeconomics
Fall 2022

1. Some policy makers have claimed that the U.S. government should purchase illegal drugs, such as cocaine, to increase the price that drug users will face. Does this idea have any merit? Illustrate this logic in a simple supply and demand framework. How does the elasticity of demand for illegal drugs relate to the efficacy of this policy? Are you more or less willing to favor this policy if you are told demand is inelastic?
2. Suppose that demand for the Honda Accord, Q_A^D , in a given year depends not only on its own price, but also by the price of the Toyota Camry and the price of gasoline. The function is

$$Q_A^D = 430 - 10P_A + 10P_C - 10P_G$$

where P_A and P_C are the prices of the Accord and the Camry, respectively, (in \$1000s), and P_G is the price of gasoline (in \$ per gallon).

- a. What is the cross price elasticity of demand for Accords with respect to the price of Camrys when the price of both cars is \$20,000 and gasoline costs \$3 per gallon?
 - b. What is the elasticity with respect to the price of gasoline (using the same information)?
 - c. Are Accords and Camrys substitutes or complements? Are Accords and gasoline substitutes or complements?
3. Read the article “The true cost to Amazon of the ‘Amazon Tax’” <http://www.chicagotribune.com/business/ct-true-cost-amazon-tax-20160114-story.html>. We will consider the effects of taxation on Amazon’s operations in some product market. Even though the supply and demand assumptions are less appropriate for many markets that Amazon operates in.
 - a. According to the article, when Amazon is forced to charge sales tax, what happens to the prices paid by buyers? The quantity of goods sold? Sales revenue to Amazon? Does this line up with the predictions of the supply and demand model?
 - b. Draw a standard supply and demand model, assume Amazon is the only seller represented by an upward-sloping supply curve. Add the tax to your model and show on your graph how the following quote about consumer expenditures could be true: “Shoppers were shelling out the same amount overall on purchases on Amazon, but ... a share of that tab was now going to taxes.”

- c. What percentage change did the tax cause in spending on Amazon products? If the tax raised prices 7.1%, what is the implied price-elasticity of consumer spending? (Hint: treat spending as if it was quantity demanded).
- d. Repeat part b for purchases over \$250. Which elasticity is higher, and does this make sense?

4. Chapter 2's Exercise 4.11, 6.10

$$2. Q_A^D = 430 - 10P_A + 10P_C - 10P_G$$

$$a. Q_A^D = 430 - 10 \times 20 + 10 \times 20 - 10 \times 3$$

$$Q_A^D = 430 - 200 + 200 - 30 = 400$$

$$\varepsilon_{AC}^D = \frac{\partial Q_A^D}{\partial P_C} \times \frac{P_C}{Q_A^D}$$

$$= 10 \times \frac{20}{400} = \frac{1}{2} \#$$

$$b. \varepsilon_{AG}^D = \frac{\partial Q_A^D}{\partial P_G} \cdot \frac{P_G}{Q_A^D} = -10 \times \frac{3}{400} = -\frac{3}{40} \#$$

c. Camrys are substitutes, gasoline is complement.

4.

$$(4.1) L_D = 200 - w$$

$$L_S = 40 + w - 2V$$

$$L_D = L_S \Rightarrow 200 - w = 40 + w - 2V$$

$$160 - 2w = -2V$$

$$80 - w = V \Rightarrow w = 80 - V \quad \frac{dw}{dV} = -1$$

$$L_D = 200 - (80 - V) = 160 + V$$

$$L_D = 200 - (80 + V)$$

$$= 120 - V$$

$$L^* = L_D$$

$$\frac{dL^*}{dV} = \frac{dL_D}{dW} \cdot \frac{dW}{dV} = -1 \times -1 = 1$$

$$\frac{dL}{dV} = -1$$

当 V 增加时 w 会减少 L* 会增加

$$(6.10) Q_D = 12 - P$$

$$Q_S = 9 + 0.5P$$

$$Q_D = Q_S \Rightarrow 12 - P = 9 + 0.5P$$

$$3 = 1.5P$$

$$2 = P \Rightarrow P^* = 2 \quad Q^* = 10$$

a. when a specific tax t

$$\Rightarrow Q_D = 12 - P - t$$

$$Q_S = 9 + 0.5P$$

$$\Rightarrow 12 - P - t = 9 + 0.5P \Rightarrow 3 - t = 1.5P$$

$$\Rightarrow P^* = 2 - \frac{t}{1.5}$$

$$2.25 - 0.75 = 1.5$$

firm receive

$$Q^* = 12 - (2 - \frac{t}{1.5}) - t$$

$$Q^* = 10 - \frac{1}{3}t$$

b. $t = 0.75 \quad 9.75 = 12 - P \Rightarrow P = 2.25$

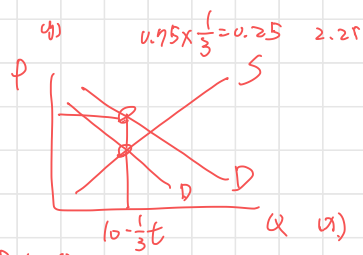
$$Q^* = 10 - \frac{1}{3} \times 0.75 = 10 - 0.25 = 9.75$$

$$\varepsilon = -1 \quad \eta = 0.5$$

$$\text{consumer tax incidence} = \frac{0.5}{0.5 - (-1)} = \frac{0.5}{1.5} = \frac{1}{3}$$

$$10 - \frac{1}{3}t = 12 - p \quad p = 2 + \frac{1}{3}t$$

$$\begin{aligned} Q_D &= 12 - p \\ Q_S &= 9 + 0.5p \end{aligned} \Rightarrow \begin{aligned} p &= 12 - Q_D \\ p &= -18 + 2Q_S \end{aligned}$$



$$p = 12 - Q_D - t$$

$$p = -18 + 2Q_S$$

$$2p = 24 - 2Q - 2t$$

$$p = -18 + 2Q$$

$$2Q = 24 - 2p - 2t$$

$$2Q = -18 - p$$

$$12 - Q - t = -18 + 2Q$$

$$3Q - t = 30$$

$$10 - \frac{1}{3}t = Q$$

$$10 - \frac{1}{3}t = 9 + 0.5p$$

$$20 - \frac{2}{3}t = 18 + p$$

$$24 - 2p - 2t = -18 - p$$

$$42 - p - 2t = 0$$

$$p = 12 - (10 - \frac{1}{3}t) - t$$

$$p = 2 - \frac{2}{3}t$$

$$p =$$

$$\begin{aligned}
 4.4 \quad Q_D &= 220 - 2P \\
 Q_S &= 20 + 3P - 20r \\
 220 - 2P &= 20 + 3P - 20r \\
 200 + 20r &= 5P \\
 \Rightarrow P &= 40 + 4r \Rightarrow \frac{dP}{dr} = 4 \\
 \frac{dQ}{dr} &= \frac{\partial Q}{\partial P} \times \frac{dP}{dr} \\
 &= -2 \times 4 \\
 &= -8
 \end{aligned}$$

$$\begin{aligned}
 5.2 \quad a. \quad Q &= 120 - 2P + 4Y \\
 P &= 10 \quad Q = 20
 \end{aligned}$$

$$\epsilon_D = \frac{\partial Q}{\partial P} \times \frac{P}{Q} = -2 \times \frac{10}{20} = -1 \#$$

$$\epsilon_D = \frac{\partial Q_C}{\partial P_C} = \frac{\partial Q/Q}{\partial P_C/P} = \frac{-3.2\%}{5\%} = \frac{4 \cdot 6.4}{5 \cdot 3^2}$$

$$b. \quad Q = 10P^{-2}$$

$$Q = 10P^{-2}$$

$$\ln Q = \ln 10 - 2 \ln P$$

$$\epsilon_D = \frac{\partial Q}{\partial P} \times \frac{P}{Q} = -20P^{-3} \times \frac{P}{Q}$$

$$= -0.64 \rightarrow \text{inelastic}$$

$$\epsilon_D = \frac{\partial \ln Q}{\partial \ln P} = -2$$

$$= -2 \cancel{P^3} \times \frac{P}{10P^{1/2}} = -2 \#$$

$$\begin{aligned}
 5.8 \quad \epsilon_{CP} &= \frac{\partial Q_C}{\partial P_P} \times \frac{P_P}{Q_C} = 16.2 \times \frac{31}{1.275} \\
 &= 0.39
 \end{aligned}$$

6.8 inelastic curve

$$Q(P(t)) \quad R = P(t) \cdot Q(P(t))$$

$$\frac{dR}{dt} = \frac{dP}{dt} Q + P \times \frac{dQ}{dP} \times \frac{dP}{dt}$$

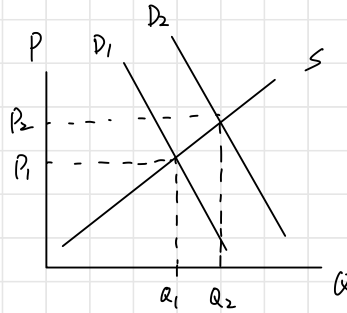
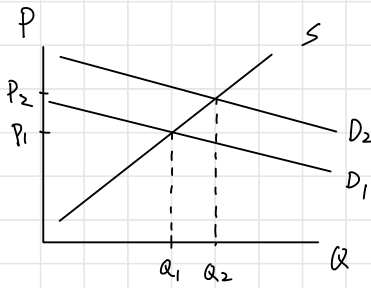
$$= \frac{dP}{dt} (Q + P \frac{dQ}{dP})$$

$$= \frac{dP}{dt} Q (1 + \frac{dQ}{dP} \times \frac{P}{Q})$$

$$= \frac{dP}{dt} Q (1 + \epsilon)$$

$$\epsilon < -1 \Rightarrow R(t) \downarrow$$

$$-1 < \epsilon < 0 \Rightarrow R(t) \uparrow$$



4.11

$$L_D = 200 - W$$

$$200 - W = 40 + W - 2V$$

$$L = 120 - V$$

$$L_S = 40 + W - 2V$$

$$2W = 160 + 2V$$

$$W = 80 + V$$

$$\frac{dL}{dV} = \frac{\partial L}{\partial W} \cdot \frac{dW}{dV} = -1 \times 1 = -1$$

$$\frac{dW}{dV} = 1$$

6.8. $p(t)$ $Q(p(t))$

$$R(t) = p(t) \cdot Q(p(t))$$

$$\frac{dR}{dt} = \frac{dp}{dt} \times Q + p \times \frac{dQ}{dp} \times \frac{dp}{dt}$$

$$= \frac{dp}{dt} \left(Q + p \times \frac{dQ}{dp} \right) \quad (\text{Q 提出})$$

$$= \frac{dp}{dt} Q \left(1 + \frac{dQ}{dp} \times \frac{p}{Q} \right)$$

$$= \frac{dp}{dt} Q (1 + \epsilon)$$

$$-1 < \epsilon < 0 \Rightarrow R(t) \uparrow$$

$$\epsilon < -1 \Rightarrow R(t) \downarrow$$